

1. Ms. Wilson invested \$10,000 in two accounts, one yielding 6% interest and the other yielding 8%. If she received a total of \$720 in interest at the end of the year, how much did she invest in each account?

The amount invested at 6% was \$.

The amount invested at 8% was \$.

2. A glass of skim milk supplies 0.1 mg of iron and 8.5 g of protein. A quarter pound of lean meat provides 3.2 mg of iron and 22 g of protein. A person on a special diet is to have 3.7 mg of iron and 64.5 g of protein. How many glasses of skim milk and how many quarter-pound servings of meat will provide this?

How many glass(es) of skim milk and how many quarter-pound serving(s) of lean meat would be required?

glass(es) of skim milk and quarter-pound serving(s) of lean meat will provide the needed nutrients.

3. Soybean meal is 12% protein; cornmeal is 6% protein. How many pounds of each should be mixed together in order to get 240-lb mixture that is 10% protein?

How many pounds of cornmeal should be in the mixture?

pounds

How many pounds of soybean meal should be in the mixture?

pounds

4. Jake has \$312,000 to invest. He chooses one money market fund that pays 6.3% and a mutual fund that has more risk but has averaged 21.3% per year. If his goal is to average 8.8% per year with minimal risk, how much should he invest in each fund?

Jake should invest \$ in the money market fund that pays 6.3%.
(Simplify your answer.)

Jake should invest \$ in the mutual fund that averaged 21.3% per year.
(Simplify your answer.)

5. A theater owner wants to divide a 2700 seat theater into three sections, with tickets costing \$50, \$80, and \$90, depending on the section. He wants to have twice as many \$50 tickets as the sum of the other tickets, and he wants to earn \$164,000 from a full house. Find how many seats he should have in each section.

The number of \$50 seats is .

The number of \$80 seats is .

The number of \$90 seats is .

- *6. An investment of \$36,000 was made by a business club. The investment was split into three parts and lasted for one year. The first part of the investment earned 8% interest, the second 6%, and the third 9%. Total interest from the investments was \$2820. The interest from the first investment was 3 times the interest from the second. Find the amounts of the three parts of the investment.

What is the amount of the first part of the investment? \$

What is the amount of the second part of the investment? \$

What is the amount of the third part of the investment? \$

7. Ace Trucking Company has an order for delivery of three products, A, B, and C. The table gives the volume in cubic feet, the weight in pounds, and the value for insurance in dollars for a unit of each of the products. If the carrier can carry 15,600 cubic feet and 16,100 pounds and is insured for \$71,600, how many units of each product can be carried?

	Product A	Product B	Product C
Volume (cu ft)	24	40	80
Weight (lb)	40	30	60
Value (\$)	150	160	300

Select the correct choice below and fill in any answer boxes within your choice.

- ☐ A. There is one solution. The company can carry units of Product A, units of Product B, and units of Product C.
- ☐ B. There are infinitely many solutions. If the company carries z units of Product C, then the company can carry units of Product A and units of Product B.
- ☐ C. There is no solution.

8. A man has \$500,000 invested in three rental properties. One property earns 7.5% per year on the investment, the second earns 8%, and the third earns 9%. The total annual earnings from the three properties is \$42,050, and the amount invested at 9% equals the sum of the first two investments. Let x equal the investment at 7.5%, y equal the investment at 8%, and z represent the investment at 9%.

Complete parts (a) through (d) below.

- a. Write an equation that represents the sum of the three investments.

(Type an equation. Do not include the \$ symbol in your answer.)

- b. Write an equation that states that the sum of the returns from all three investments is \$42,050.

(Type an equation. Do not include the \$ symbol in your answer.)

- c. Write an equation that states that the amount invested at 9% equals the sum of the other two investments.

(Type an equation. Do not include the \$ symbol in your answer.)

- d. Solve the system of equations to find how much is invested in each property.

- ☐ A. There is one solution. The man invested \$ at 7.5%, \$ at 8%, and \$ at 9%.
- ☐ B. There are infinitely many solutions. If the man invests \$ z at 9%, then he invests \$ at 7.5% and \$ at 8%.
(Type expressions using z as the variable.)
- ☐ C. There is no solution.

9. Ben was in charge of ordering 30 pizzas for the office party. He ordered three types of pizza: Cheese, Pepperoni, and Supreme. The cheese pizzas cost \$7 each, the pepperoni pizzas cost \$10 each, and the supreme pizzas cost \$13 each. He spent exactly twice as much on the pepperoni pizzas as he did on the cheese pizzas. If Ben spent a total of \$288 on pizza, how many pizzas of each type did he buy?

Ben ordered cheese pizzas, pepperoni pizzas, and supreme pizzas.

10. The nutritional content per ounce of three foods is presented in the table below. If a meal consisting of the three foods allows exactly 4500 calories, 123 grams of protein, and 9150 milligrams of vitamin C, how many ounces of each kind of food should be used?

	Calories	Protein (in grams)	Vitamin C (in milligrams)
Food A	100	7	350
Food B	500	13	1800
Food C	500	9	50

Food A: oz

Food B: oz

Food C: oz